JEE NEXUS - NCERT CLASS 12 - PHYSICS

Chapter 3: Current Electricity

NCERT: Current Electricity | Avg PYQs: 2 to 3 Questions per Shift | Weightage: 8-12 Marks | D<=>: Moderate

*** HIGH REPEAT CHAPTER (MUST DO)**

HIGH-YIELD JEE SUBTOPICS:

Kirchhoff's Laws & Wheatstone Bridge - Drift Velocity & Mobility Relations - Cell Combinations & EMF vs Terminal Potential - Galvanometer to Ammeter/Voltmeter Conversion - Meter Bridge & Temperature Dependence of Resistivity

NTA Focus: Direct numerical value questions on circuit loop currents and galvanometer coil resistance calculations.

1. Microscopic Ohm's Law & Drift Mechanics

In conductors, free electrons execute random thermal motion with velocities around 10^5 m/s, but net thermal current is zero due to isotropic velocity distribution.

Upon applying an external electric field E, electrons experience acceleration -eE/m and acquire an average steady drift velocity v_d opposing the electric field between successive relaxation collisions.

MOST REPEATED PYQ TOPIC (80% JEE Questions)

NTA Favorite: Drift Velocity & Current Density Relations

Always remember the link between microscopic and macroscopic quantities. Current I through a conductor of cross-section A is given by:

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I = n e A v_d    ==>    J = (I / A) = n e v_d = sigma E = (E / rho)
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- Drift velocity magnitude: v_d = (e E tau) / m
- Mobility u = v_d / E = (e tau) / m (Always positive, SI unit: m²/(V·s))
- Conductivity sigma = n e² tau / m and Resistivity Á = m / (n e² tau)

COMMON JEE TRAP / SILLY MISTAKE ALERT

Common Student Trap: Stretching a Wire vs Cutting

When a wire of initial resistance R is STRETCHED by n times its length (or radius halved), VOLU
alone!

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R = rho (L / A) = rho (L^2 / Volume) ==> R_{new} = n^2 R_{initial}
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- If stretched so length increases by x% (where x < 5%): Delta R/R ~= 2(Delta L/L)
- If stretched so radius decreases by 1%: Delta R/R ~= -4(Delta r/r) = +4% increase

2. Kirchhoff's Laws & Wheatstone Bridge Archetypes

Kirchhoff's Current Law (KCL) is based on the Conservation of Charge (Sum I_in = Sum I_out). Kirchhoff's Voltage Law (KVL) is based on the Conservation of Energy (Sum Delta V = 0 around any closed mesh).

A Wheatstone bridge consists of four resistors. When the galvanometer shows zero deflection (null point), the bridge is balanced and no current flows through the central branch.

FORMULA & SHORTCUT VAULT

Master Formula Vault: Wheatstone Condition & Maximum Sensitivity

For null deflection in bridge ABCD with resistances P, Q, R, S:

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(P / Q) = (R / S)    <==>    V_B = V_D
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- Galvanometer and battery can be interchanged without affecting balance condition.
- The bridge is most sensitive when all four resistances P, Q, R, S are of approximately equal magnitude.

DIRECT NCERT STATEMENT / EXCEPTION

Direct NCERT Statement: Carbon Resistor Color Code & Tolerances

NCERT Table 3.2: "BBROY Great Britain Very Good Wife" - Black (0), Brown (1), Red (2), Orange (3), Yellow (4), Green (5), Blue (6), Violet (7), Grey (8), White (9). Gold (+/-5%), Silver (+/-10%), No Color (+/-20%).

METHOD / DERIVATION: Step-by-Step Nodal Analysis for Complex Multi-Loop Circuits

[Step 1] Identify all essential nodes and assign one node as Reference/Ground (V = 0 V).

[Step 2] Assign unknown node potentials V1, V2 to the remaining nodes.

[Step 3] Write KCL at each node: sum of outward currents = 0: Sum (V_node - V_neighbor) / R = 0.

[Step 4] Solve linear algebraic equations directly using substitution or Cramer's Rule.

3. Cells, Combinations & Meter Bridge

An electrochemical cell maintains a potential d<=>erence between its electrodes. The EMF E is the potential d<=>erence between terminals in an open circuit (zero current).

When drawing current I, terminal potential V = E - I·r due to internal resistance r. During charging by an external charger, V = E + I·r.

FORMULA & SHORTCUT VAULT

Master Formula Vault: Mixed Cell Combinations & Max Power

For n identical cells in m parallel rows with external load R:

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I = (n E / R + frac{n r){m}},    P_{max} = (n m E^2 / 4 r)    when  R = (n r / m)
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- Parallel cells of unequal EMF: E_eq = (E1/r1 + E2/r2) / (1/r1 + 1/r2)
- Equivalent internal resistance: 1/r_eq = 1/r1 + 1/r2

MOST REPEATED PYQ TOPIC (80% JEE Questions)

NTA Favorite: Galvanometer Conversion to Ammeter & Voltmeter

1. Ammeter: Connect small shunt resistance S in PARALLEL with galvanometer G: S = (I_g · G) / (I - I_g)
2. Voltmeter: Connect high resistance R in SERIES with galvanometer G: R = (V / I_g) - G

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S = (I_g G / I - I_g)    and    R = (V / I_g) - G
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- Ideal Ammeter resistance = 0 Ohm. Ideal Voltmeter resistance = " Ohm.
- If range is increased n times (I = n I_g): S = G / (n - 1)
- If voltmeter range is increased n times (V = n V_g): R = (n - 1) G

Top Solved PYQ Archetypes (2)

Archetype #1 (JEE Main 2024 / 2023 Repeating Archetype)

Concept: Galvanometer to Multi-Range Ammeter & Shunt Calculation

A galvanometer with coil resistance 50 Ohm gives full-scale deflection for a current of 10 mA. To convert this galvanometer into an ammeter reading up to 5 A, find the value of shunt resistance required.

Methodical Solution Steps:

- Given: Galvanometer resistance G = 50 Ohm, Full-scale deflection current I_g = 10 mA = 0.01 A.
- Total current to be measured I = 5 A.
- Current passing through shunt S: I_s = I - I_g = 5 - 0.01 = 4.99 A.
- Since shunt and galvanometer are in parallel, V_s = V_g: S · (I - I_g) = I_g · G.
- S = (0.01 × 50) / 4.99 = 0.50 / 4.99 ~= 0.1002 Ohm.

NTA Trap & Shortcut:

Shortcut: S = G / (n - 1) where n = I / I_g = 5000 / 10 = 500. S = 50 / 499 ~= 0.1002 Ohm. Beware of units: 10 mA must be converted to Amperes!

Final Answer: 0.10 Ohm (approx)

Archetype #2 (JEE Main 2023 / 2021 Numerical)

Concept: Unequal Cells in Parallel with Internal Resistance

Two cells of EMF 2 V and 4 V with internal resistances 1 Ohm and 2 Ohm respectively are connected in parallel with their like terminals together. Calculate the equivalent EMF and terminal potential across an external load of 5 Ohm.

Methodical Solution Steps:

- Formula for two parallel cells: E_eq = (E1/r1 + E2/r2) / (1/r1 + 1/r2)
- E_eq = (2/1 + 4/2) / (1/1 + 1/2) = (2 + 2) / (1.5) = 4 / 1.5 = 8/3 V ~= 2.67 V.
- Internal resistance r_eq: 1/r_eq = 1/1 + 1/2 = 3/2 => r_eq = 2/3 Ohm.
- Circuit current through 5 Ohm load: I = E_eq / (R + r_eq) = (8/3) / (5 + 2/3) = (8/3) / (17/3) = 8/17 A.
- Terminal potential V = I × R = (8/17) × 5 = 40/17 V ~= 2.35 V.

NTA Trap & Shortcut:
Common mistake: Students directly average EMFs (2 + 4)/2 = 3 V, which ignores internal resistance weighting!

Final Answer: E_eq = 2.67 V (8/3 V); V = 2.35 V (40/17 V)

NCERT Hidden Gems & Tables (2)

Manganin and Constantan for Standard Resistors (NCERT Class 12 Part 1, Page 104)

"NCERT Section 3.7: Alloys such as Manganin, Constantan, and Nichrome are used for standard wire-wound resistors because their resistivity is very high and has extremely low temperature coefficient of resistance (almost independent of temperature)."

Why NTA Asks This: Asked directly as a statement/assertion-reason question in JEE Main 2022 & 2024.

Semiconductor vs Metallic Resistivity Curve (NCERT Class 12 Part 1, Page 105 (Fig 3.9 & 3.10))

*"For metals, resistivity increases with temperature (positive alpha). For semiconductors and insulators, resistivity DECREASES exponentially with increasing temperature because charge carrier density n increases exponentially: n(T) " exp(-E_*

Why NTA Asks This: NTA frequently asks students to match the correct Á vs T graph for Copper, Nichrome, a

Checklist & Personal Notes

Must-Solve Subtopic Checklist:

- [X] Derive Drift Velocity v_d = eEtau/m and Current Relation I = n e A v_d
- [X] Solve Balanced & Unbalanced Wheatstone Bridge with Symmetrical Fold Method
- [ ] Calculate Terminal Voltage during Charging vs Discharging: V = E + Ir vs V = E - Ir
- [ ] Convert Galvanometer (G) to Ammeter of range I and Voltmeter of range V
- [ ] Calculate Maximum Power Transfer across variable load (R = r_eq)
- [ ] Analyze temperature coefficient of resistance R(T) = R_0[1 + alpha(T - T_0)]

My Chapter Scratchpad / Student Notes:

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Remember shunt is parallel, multiplier resistor is in series. V = E - Ir.
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